

Maximizing Player Satisfaction: Collaborative Strategies in Two-Player Gnomes Game

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Introduction

Users	Legislators	Manufacturers
☐ Safety	☐ Speed Limits	☐ Fuel efficiency
☐ Time	☐ Abiding Laws	☐ Durability

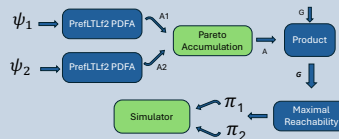


- ❖ Stakeholders have different preferences about various attributes.
- ❖ Autonomy must determine the best trade-off to satisfy all stakeholders.
- ❖ We use game theory to determine the best trade-offs.

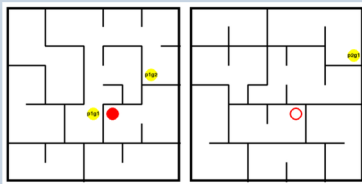
Research Goal

Develop and implement a provably-correct algorithm to synthesize cooperative strategies in a two-player games where both players aim to maximally satisfy their (possibly incomplete) preferences.

Methods



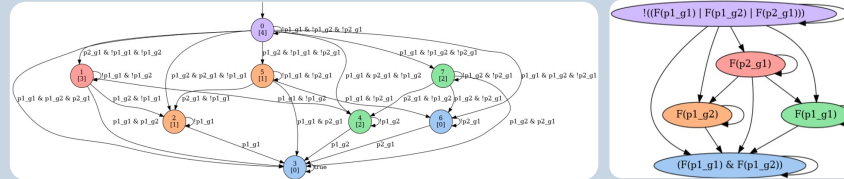
Inputs: 2 sets of preferences and a Gnomes game model
Outputs: Strategies for simulator



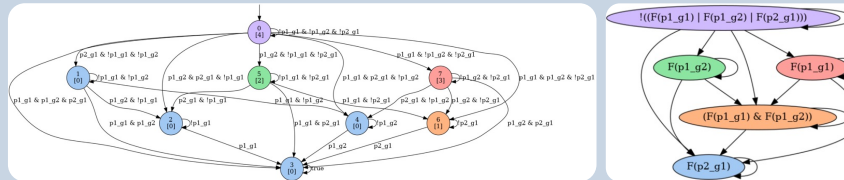
Player 1 Preferences Ranked
 $(\Diamond p1g1 \triangleright \Diamond p2g1) \& (\Diamond p1g2 \triangleright \Diamond p2g1)$
Player 2 Preferences Ranked
 $(\Diamond p2g1 \triangleright \Diamond p1g1) \& (\Diamond p2g1 \triangleright \Diamond p1g2)$

Figures and Results

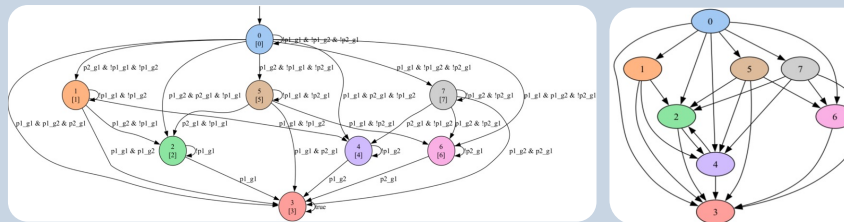
Player 1 Preferences



Player 2 Preferences



Pareto Results



Conclusion

- ❖ Insert the strategies into the simulator
- ❖ Check the state of the token after 10 turns
- ❖ Perform the same experiment with a human to determine the logical position of the token
- ❖ Compare the results to evaluate the effectiveness of our algorithm

Future Work

- ❖ Expand Pareto accumulation to multi-player games, examining complexity and group dynamics.
- ❖ Movement constraints: Limit players to visit each state only once.
- ❖ Investigate strategic deception, such as motivations for lying similar to those in Werewolf.
- ❖ Apply Pareto principles to real-world applications in economics, politics, or autonomous vehicle strategies.

Acknowledgments

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References

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- ❖ Jorge, A., and McIlraith, S. A. "Planning with preferences." AI Magazine 29.4 (2008): 25-25.
- ❖ Shenghui C., Fried D., and Topcu U. "Human-Agent Cooperation in Games under Incomplete Information through Natural Language Communication." arXiv preprint arXiv:2405.14173 (2024).